

Investigating Expert Collapse in EveNet Moe

Overview

- Simple dataset (MNIST)
- Baseline (non-MoE) transformer model
- Simple MoE implementation
- Isolated EveNet MoE implementation

Baseline Transformer (ViT)

- Single-layer Vision Transformer on 28×28 images, patched to 14×14
- CLS token + learnable positional embeddings
- Self-attention $\rightarrow$ LayerNorm $\rightarrow$ **standard FFN MLP** $\rightarrow$ LayerNorm $\rightarrow$ linear classifier
- FFN: two Linear layers (hidden dim 128, ReLU) - every token passes through the same network

MoE Recap

- Core idea: dynamically routes inputs to specialized subnetworks ("experts")
- Each expert is a subnetwork (the number of experts is a hyperparameter)
 - In a transformer architecture they replace the feedforward layer
- Purpose is to allow specialization on different input features
- A gating network determines which experts to activate for each input
- Only a subset of the experts are activated for each input (hyperparameter)
- Typically implemented as a classifier

Simple MoE Transformer

- Same backbone - FFN replaced with a **4-expert Mixture-of-Experts** layer
- Each expert: hidden dim 32 (total capacity matches baseline FFN)
- Top-2 hard: linear gate scores tokens, dispatches to 2 highest-scoring experts
- Tokens processed independently by their selected experts, recombined with softmax weights
- Single load-balancing loss term (coefficient 0.05) to encourage even utilisation

EveNet MoE Main Upgrades

- 8 routed experts + 0 shared experts - twice the specialisation of simple MoE
- Each expert: hidden dim 64, GELU activation (matches EveNet)
- Top-2 routing with stochastic noise: extra learned noise head keeps exploration alive during training
- Two balancing terms: auxiliary loss ($\alpha = 0.01$) + logit-magnitude regularisation ($c_z = 0.001$)

Training Setup

Hyperparameter	Value
Dataset	MNIST (60K train / 10K test)
Embedding dim	64
Epochs	25
Batch size	64
Optimizer	Adam, LR 0.001 + cosine annealing
Device	CUDA

Both MoE variants trained from scratch, independently of each other and the baseline.

Performance Comparison

Metric	Baseline	Simple MoE	EveNet MoE
Test Accuracy	96.52 %	92.91 % (−3.6 pp)	96.61 % (+0.1 pp)
Test Loss	0.124	0.247 (2×)	0.150 (1.2×)
Final Train Loss	0.347	-	0.024

- Simple MoE underperforms badly - EveNet MoE **matches baseline accuracy** despite only activating 2 of 8 experts per token

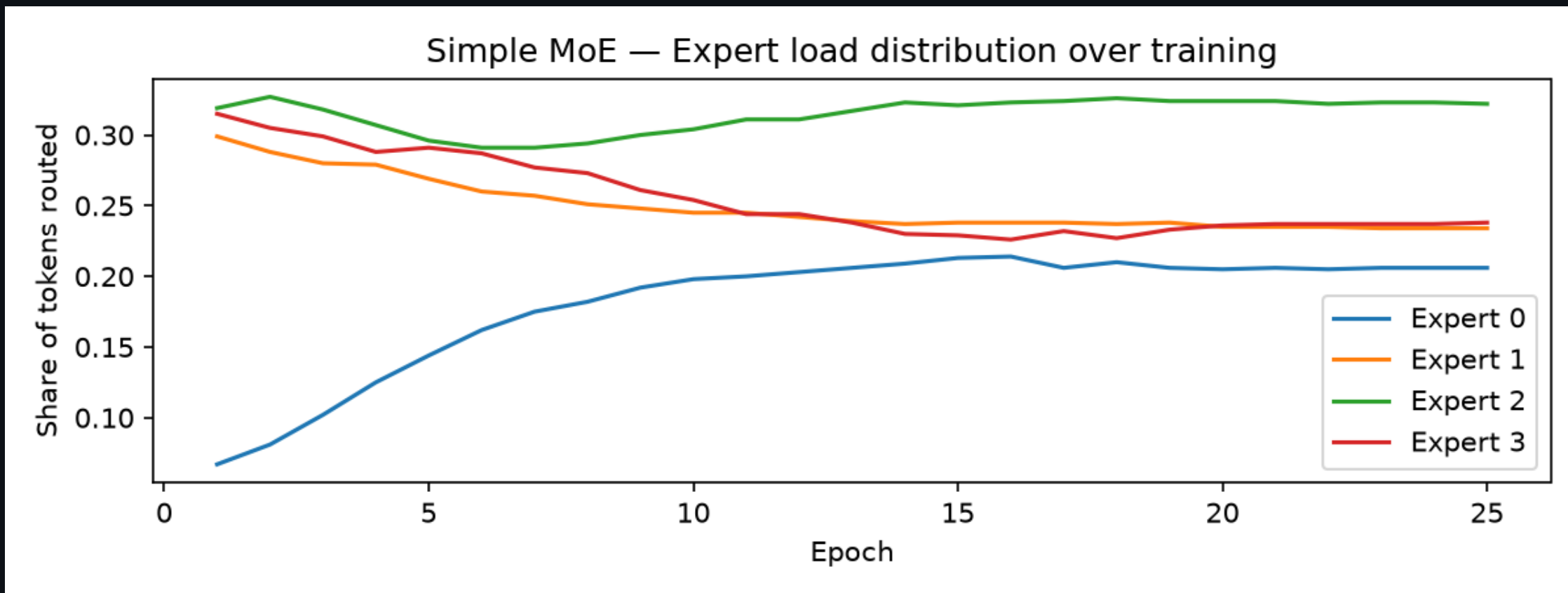
Expert Routing Distribution (Final Epoch) - Simple MoE

Expert	Share	Deviation
E0	20.6 %	−4.4 pp
E1	23.8 %	−1.2 pp
E2	32.2 %	+7.2 pp
E3	23.8 %	−1.2 pp

Expert Routing Distribution (Final Epoch) - EveNet MoE

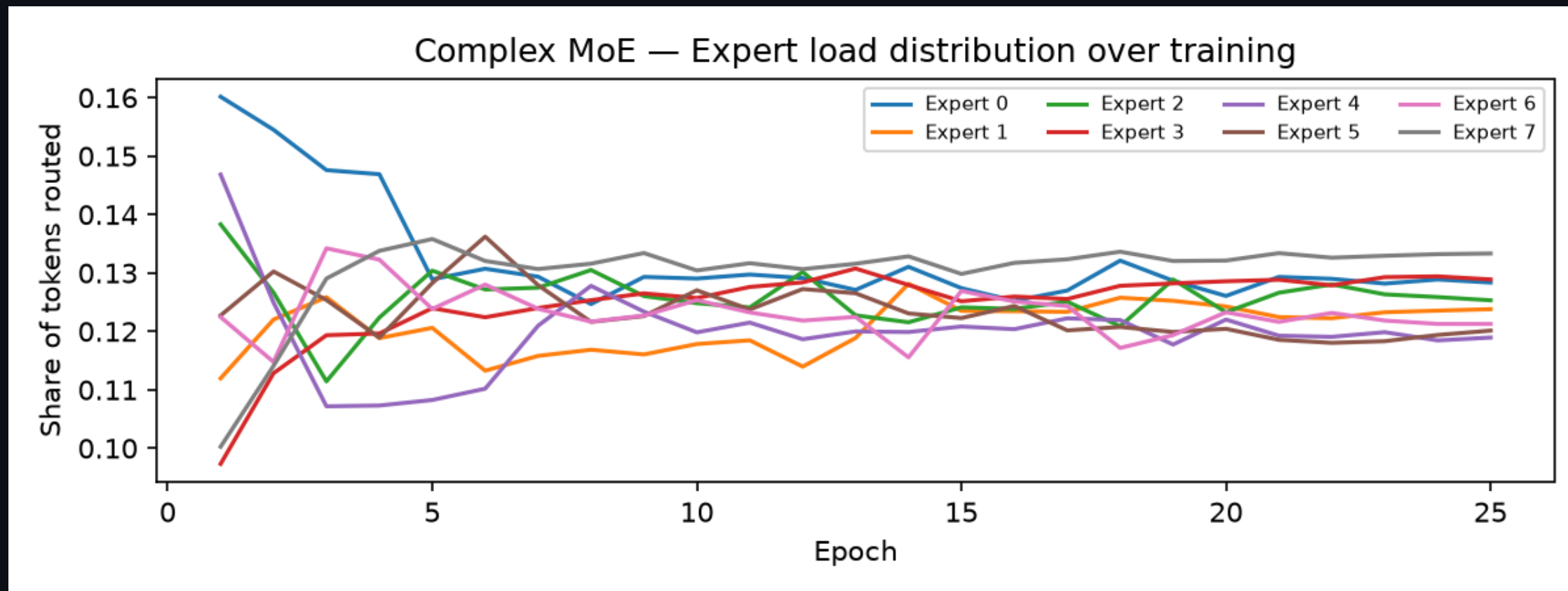
Expert	Share	Deviation
E0	12.4 %	−0.1 pp
E1–E6	12.5–12.7 %	≤ +0.2 pp
E7	11.8 %	−0.7 pp

Simple MoE - Routing Evolution



- Expert 0 starts at 6.7 %, only reaches 20.6 % after 25 epochs
- **Expert 2 absorbs ~32 %**

EveNet MoE - Routing Evolution



- **Nearly uniform from epoch 4 onward** - all experts within ± 1.5 pp of fair share

Key Takeaways

- **Simple MoE shows mild expert collapse:** E0 never recovers from a cold start (6.7 % → 20.6 %), E2 dominates
- **EveNet MoE achieves near-uniform routing** within 4 epochs and maintains it - max deviation only -0.7 pp
- The combination of **stochastic noise + dual balancing losses** is what makes the difference